Name	Index Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018

BIOLOGY **9744/01**

HIGHER 2

14 SEPTEMBER 2018
Fri 0800 – 0900

1 hour

Paper 1 Multiple Choice

Additional material: Optical Mark Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your NRIC number, name and CTG on the Optical Mark Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.

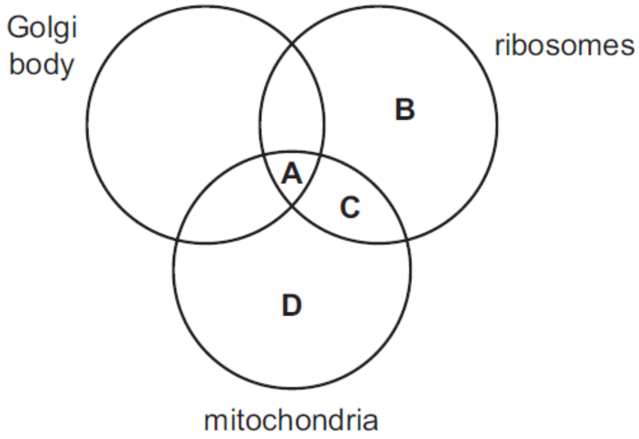
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

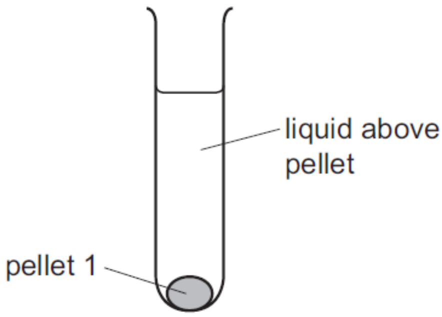
This question paper consists of **23** printed pages and **1** blank page.

1 *Plasmodium* is a genus of unicellular eukaryotes that are obligate parasites.

Which of these cell structures are present in *Plasmodium*?



2 A scientist carried out an experiment to separate cell structures in animal cells. The cells were broken open to release the cell structures. This extract was filtered into a centrifuge tube and then spun in a centrifuge. The heaviest cell structure sank to the bottom forming pellet 1, as shown in the diagram.



The liquid above pellet 1 was poured into a clean centrifuge tube and spun in the centrifuge at a higher speed to separate the next heaviest cell structure. This cell structure sank to the bottom, forming pellet 2.

This procedure was repeated twice more to obtain pellet 3 and pellet 4, each containing a single type of cell structure.

Which row shows the order in which the cell structures were collected?

	Pellet 1	Pellet 2	Pellet 3	Pellet 4
A	Nucleus	Lysosomes	Mitochondria	Ribosomes
B	Nucleus	Mitochondria	Lysosomes	Ribosomes
C	Ribosomes	Lysosomes	Mitochondria	Nucleus
D	Ribosomes	Mitochondria	Lysosomes	Nucleus

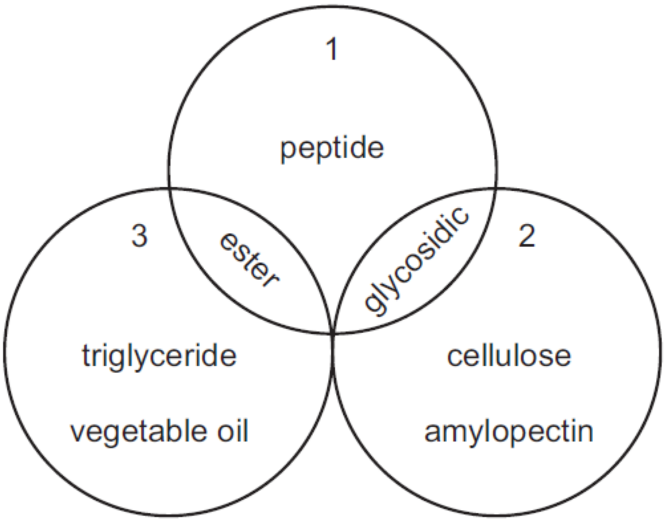
3 Some features of haemoglobin are listed.

1. They are made up of a protein component called globin and a non-protein component called haem group.
2. There are two types of polypeptide chains: the α-helix globin chain and the β-globin chain.
3. Each polypeptide chain is held by hydrophobic interactions, hydrogen and ionic bonds and disulfide bonds.
4. Most of its hydrophilic polar amino acid residues are on the external surface of the globular structure.

Which of these statements are true?

A 1 and 3 **B** 1 and 2 **C** 1, 2 and 4 **D** 1 and 4

4 The diagram shows relationships between some important molecules and bonds.



What is represented by circles numbered 1, 2 and 3?

	1	2	3
A	bonds formed by condensation	Lipids	Carbohydrates
B	bonds formed by condensation	Carbohydrates	Lipids
C	bonds formed by hydrolysis	Carbohydrates	Lipids
D	bonds formed by hydrolysis	Lipids	Carbohydrates

5 Which statements about the cell surface membrane are correct?

1 Channel proteins allow water-soluble ions and molecules across the membrane.

2 Glucose can pass into the cell via carrier proteins.

3 Oxygen passes freely through the membrane as it is soluble in lipids.

4 Some glycoproteins act as antigens.

A 1, 2, 3 and 4

B 1, 3 and 4 only

C 1 and 2 only

D 2, 3 and 4 only

6 A student investigated the hydrolysis of lipids in high-fat milk, using the enzyme lipase.

- 1 cm^3 of enzyme solution was added to 10 cm^3 of high-fat milk.
- The temperature was kept constant.
- The pH of the reaction mixture was recorded at time 0 minutes and every minute for 20 minutes.

Which statements could be supported by the results of the investigation?

1 Less product is made as time proceeds because the substrate is decreasing.

2 The pH of the reaction mixture changes more rapidly in the first few minutes and then changes less rapidly.

3 The product gradually causes more lipase molecules to denature.

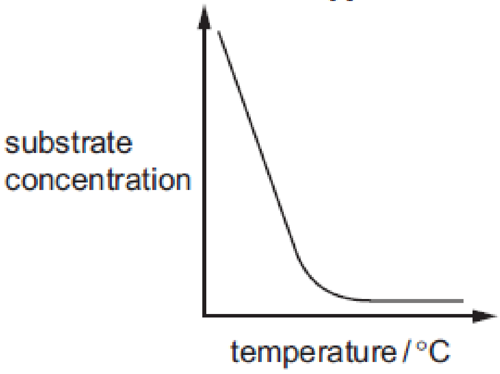
A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

7 A student carried out an investigation into the effect of temperature on the rate of an enzyme-catalysed reaction.

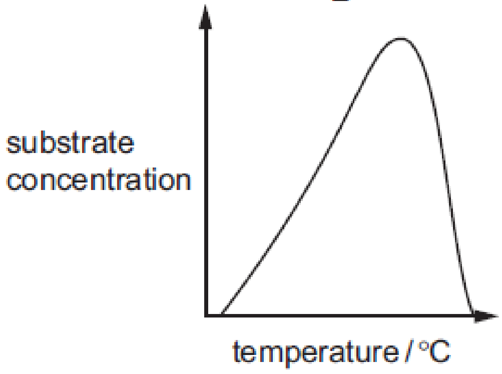
At each temperature, the substrate concentration was measured after 10 minutes. All the other variables were kept constant.

Which graph shows the effect of increasing temperature on the substrate concentration after 10 minutes?

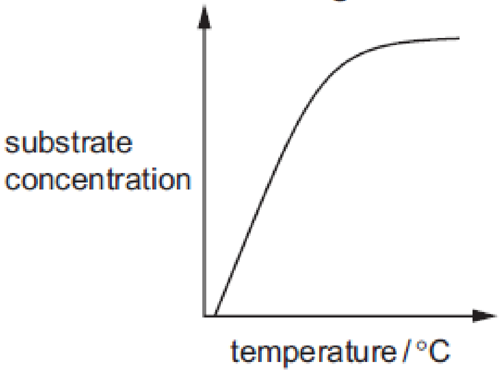
A



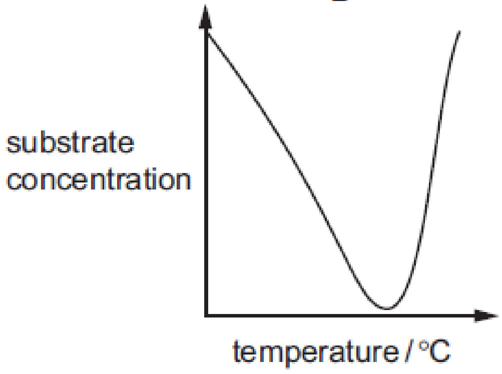
B



C



D



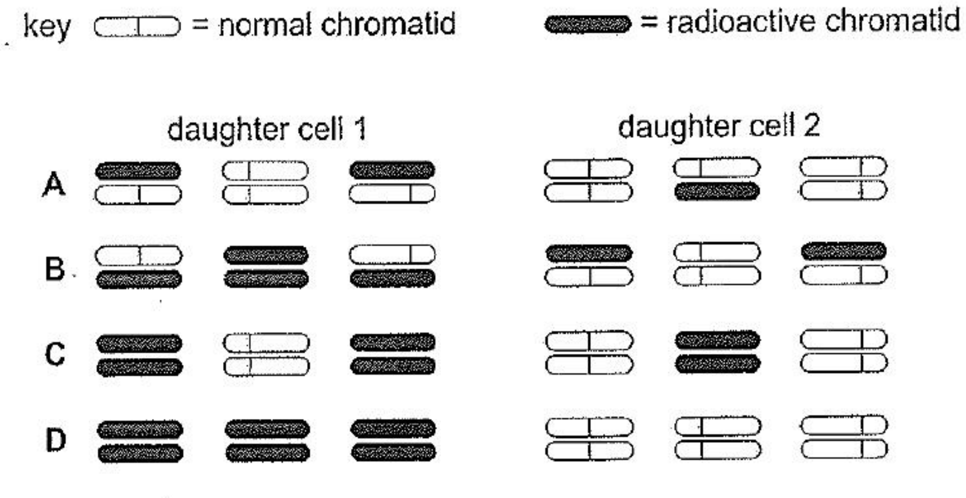
8 The following experiment was carried out.

1 Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive ^{15}N, so that all the DNA was labelled.

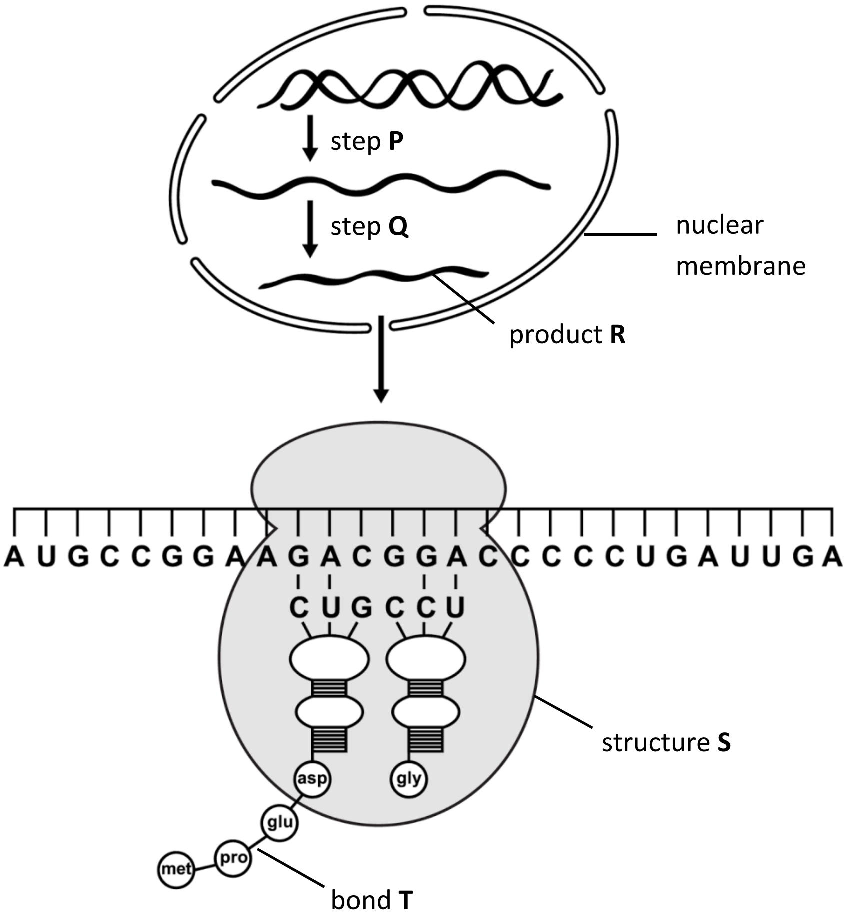
2 Cells in early interphase were then transferred to a medium with normal ^{14}N.

3 A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?



9 The diagram outlines the production of protein in a cell.



W
Which of the following statements are correct?

1 Bond T is catalysed by tRNA found in structure S.

2 RNA polymerase is active during step P and removal of exons by spliceosome occurs at step Q to give product R.

3 The mRNA shown will code for a protein containing eight amino acids.

4 The template DNA strand involved has the base sequence TACGGCCTTCTGCCTGGGGGACTAACT.

A 1 and 4

B 3 and 4

C 1, 2 and 3

D 1, 3 and 4

10 Which of these statements about cytokinesis is always true?

1 Cell structures replicate.

2 Cell structures are shared between two cells.

3 Nuclear envelope reforms.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 only **D** 3 only

11 Which of the following statements about obtaining human embryonic stem cells for research is correct?

A Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.

B The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.

C The cells must be removed within a day following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.

D The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.

12 Which of the following statements correctly describes the effect of carcinogens on lung tissue that causes a tumour to develop?

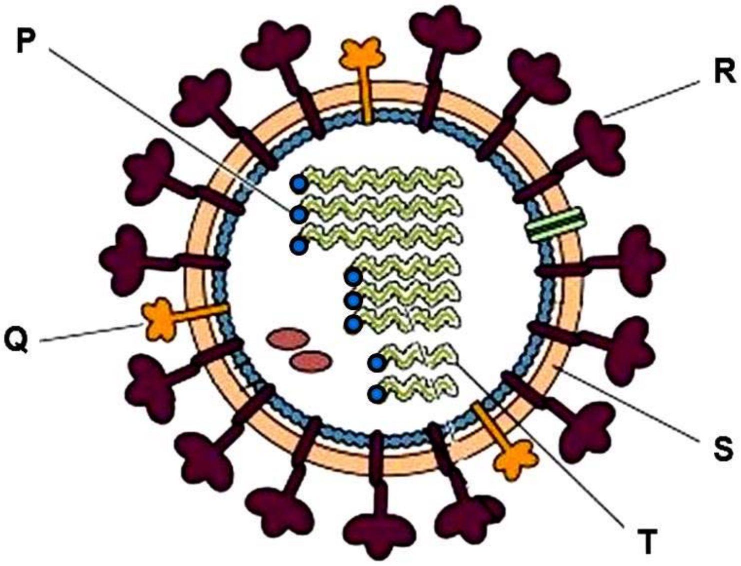
A Cells in damaged alveoli walls divide more rapidly to replace damaged areas.

B Cilia are paralysed and mucus accumulates in the lungs causing DNA to change.

C DNA changes, causing bronchial epithelial cells to divide by mitosis in an uncontrolled manner.

D Haemoglobin carries less oxygen, causing bronchial cells to divide by mitosis in an uncontrolled way.

13 The diagram shows the structure of an influenza virus.



Which of the following statements concerning the lettered components are correct?

1 Mutations that disrupt the function of P will result in the inability of the virus to initiate infection in the host cell.

2 Influenza viruses are able to cause pandemics in humans due to changes in Q and R by a slight mutation.

3 Q and R are first synthesized, and embedded in the host cell surface membrane before budding takes place.

4 S is synthesized in the host cell using the host cell enzymes before assembly of the virus.

5 Different T from influenza strains infecting different animals can be assembled into one virus within one organism.

A 1, 2 and 5

B 1, 3 and 5

C 2, 3 and 4

D 3, 4 and 5

14 What of the following statements about translation in prokaryotes is correct?

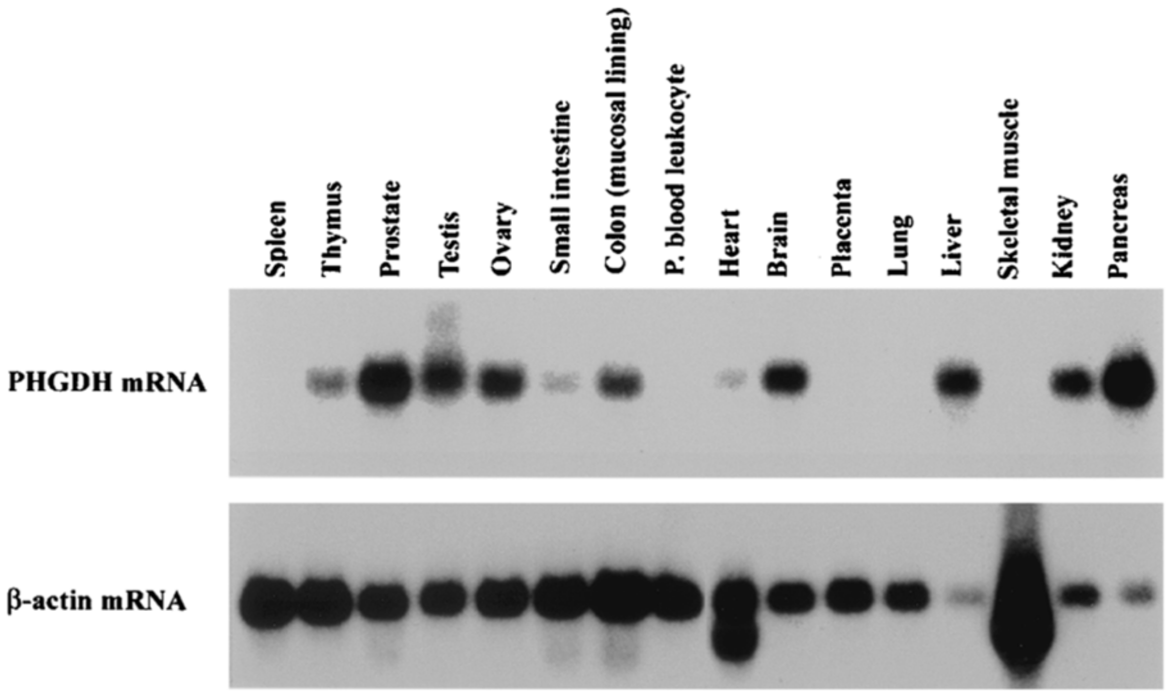
A mRNA is modified at the 5' leading end after transcription, with a 5' cap and a methylated guanosine triphosphate added to initiate translation by binding the mRNA.

B The small ribosome sub unit binds to the Shine-Dalgarno sequence on the mRNA which is the ribosome binding site.

C The tRNA with methionine binds to the small ribosomal subunit and resides in the ribosome's P site.

D There is one release factor to terminate polypeptide synthesis by binding to the different stop codons on mRNA.

15 Both β-actin and D-3-phosphoglycerate dehydrogenase (PHGDH) are proteins that can be found in the human body. Multiple tissues from the same individual were taken, followed by the isolation of the respective mRNA from the same number of cells of each tissue type. The mRNA were then subjected to gel electrophoresis. The following autoradiograph shows the result of this study.



With reference to the diagram above, which of the following statements is **not** true?

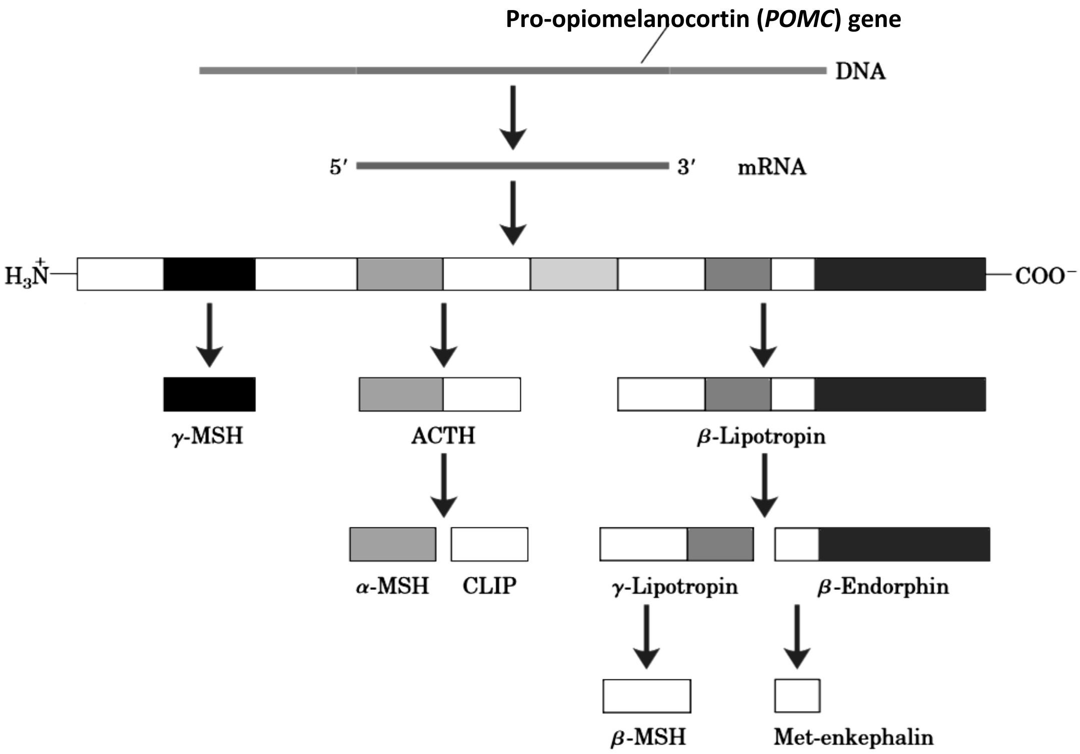
A Activators are bound to enhancers of the gene coding for β-actin in skeletal muscle cells.

B Repressors are bound to silencers of the gene coding for PHGDH in lung cells.

C The gene coding for β-actin is located in euchromatin.

D The gene coding for PHGDH cannot be found in all types of cells.

16 The pro-opiomelanocortin (*POMC*) gene is expressed in the pituitary gland, the hypothalamus, the skin and the reproductive organs. This gene codes for a 285-amino acid polypeptide that undergoes processing to form nine different peptide hormones as shown in the schematic diagram below.



The processing of POMC polypeptide involves extensive proteolytic cleavage at sites shown to contain regions of basic amino acid sequences. The proteases that recognise these cleavage sites are tissue-specific.

Multiple hormones are thus produced such as adrenocorticotropic hormone (ACTH) and β-lipotropin in the anterior pituitary under the stimulation of corticotropin releasing hormone, as well as γ-lipotropin and β-endorphin in the intermediate lobe of pituitary gland under the stimulation of dopamine.

The terminal residues of these peptide hormones are often glycosylated or acetylated.

Which of the following statements is correct?

A Specific chemical signals are required for the formation of unique peptide hormones in different tissues

B The types of hormones formed in a specific tissue depend on the control of gene expression at the translational level.

C The different hormones are formed by alternative splicing.

D Parts of the amino acid sequence of POMC polypeptide can undergo rearrangement to form peptide hormones of varying length.

17 Which statement describes a common feature of the centromere and the telomere?

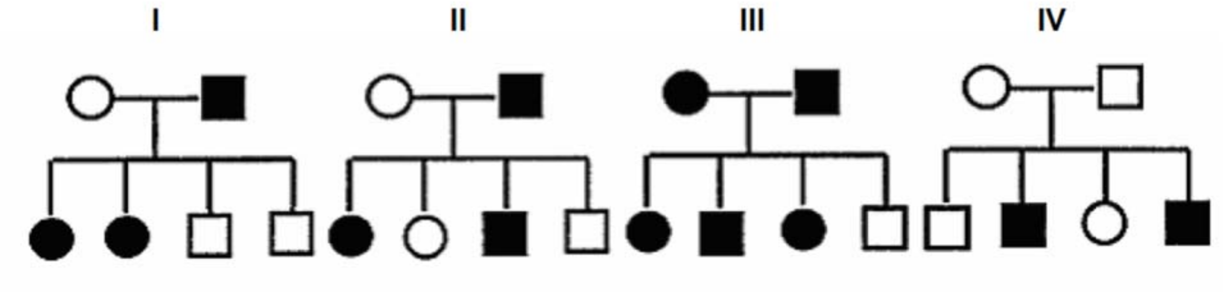
A Both are made up of DNA with a high incidence of adenine and thymine.

B Both areas undergo DNA replication simultaneously during interphase.

C Both areas have proteins associated with them.

D Both have DNA sequences that are conserved throughout the life of the organism.

18 The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male.



Which pedigree(s) show(s) a sex-linked dominant trait in humans?

A I only

B I and III

C II and III

D III and IV

19 Sucrose and maltose are disaccharides. Sucrose contains a glucose and a fructose while maltose contains two glucose units. Once broken down to monosaccharides, glucose being the first respiratory substrate, will enter glycolysis.

A student is investigating whether there is a significant difference in time taken by yeast cells to break down sucrose and maltose. A *t*-test is used to compare two sets of data to establish if they are statistically different. The $T_{calculated}$ value is 3.73.

Table 19.1

Number of data	Type of carbohydrate
8	Sucrose
8	Maltose

Table 19.2

Two Tailed Significance						
Degrees of freedom	$\alpha = 0.20$	0.10	0.05	0.02	0.01	0.002
1	3.078	6.314	12.706	31.821	63.657	318.300
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.305	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733

Using the information above, which of the following statements is correct?

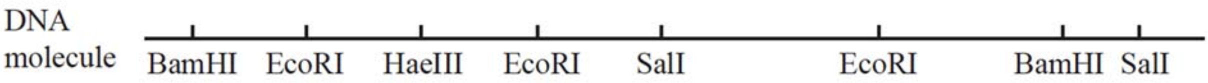
A $T_{calculated}$ value is 3.73 is higher than $T_{critical}$ value of 2.131, hence it is statistically insignificant.

B $T_{calculated}$ value is 3.73 is higher than $T_{critical}$ value of 2.131, hence it is statistically significant.

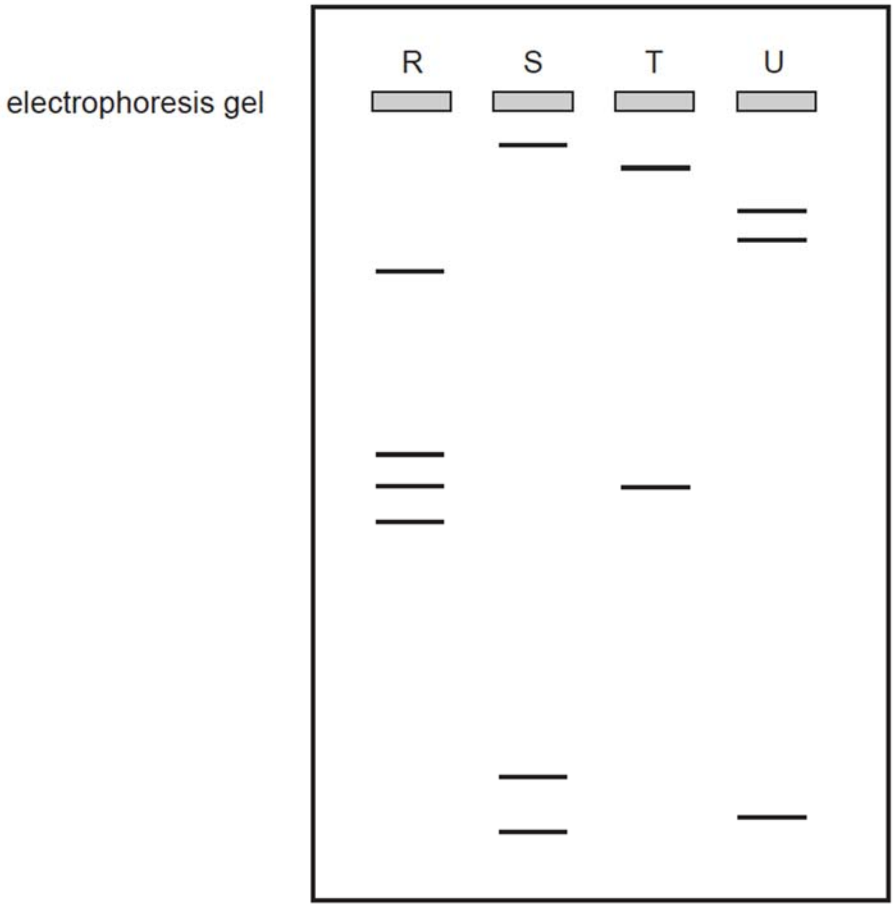
C $T_{calculated}$ value is 3.73 is higher than $T_{critical}$ value of 2.145, hence it is statistically significant.

D $T_{calculated}$ value is 3.73 is higher than $T_{critical}$ value of 4.303, hence it is statistically significant.

20 The diagram below represents a DNA molecule and the position of the recognition sites for the restriction enzymes *Bam*HI, *Eco*RI, *Hae*III and *Sal*I.



The electrophoresis gel below shows the separation of DNA segments resulting from digestion of the DNA molecule with one of the restriction enzymes.

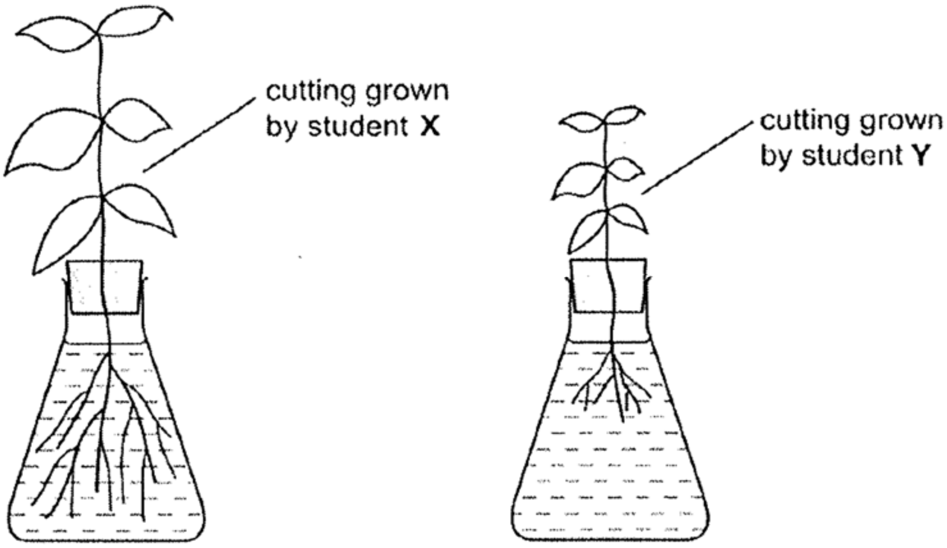


Which of the following shows the correct match between the lane and the restriction enzyme used to digest the DNA molecule?

	R	S	T	U
A	*Eco*RI	*Bam*HI	*Hae*III	*Sal*I
B	*Eco*RI	*Bam*HI	*Sal*I	*Hae*III
C	*Hae*III	*Sal*I	*Bam*HI	*Eco*RI
D	*Sal*I	*Eco*RI	*Hae*III	*Bam*HI

21 A student, X, looked after a plant. Another student, Y, looked after another plant of the same species. Each student followed the same instructions to set up the apparatus to take cuttings from their plant and grow the cuttings next to the plant from which the cuttings had been taken.

The results after one week, are shown below.



Which factors could have caused the different appearance of the two cuttings?

1 Genetically different cuttings.

2 Genotypic variation due to environment.

3 Mutation due to environment.

4 Phenotypic variation due to environment.

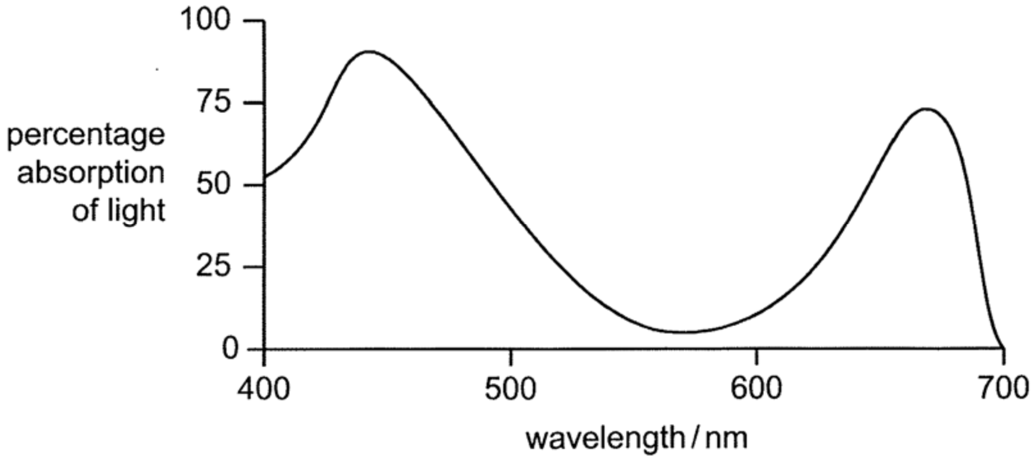
A 1 and 4

B 2 and 3

C 2 and 4

D 1, 3 and 4

22 The graph shows the absorption of light at different wavelengths by intact chloroplasts from a pond weed.



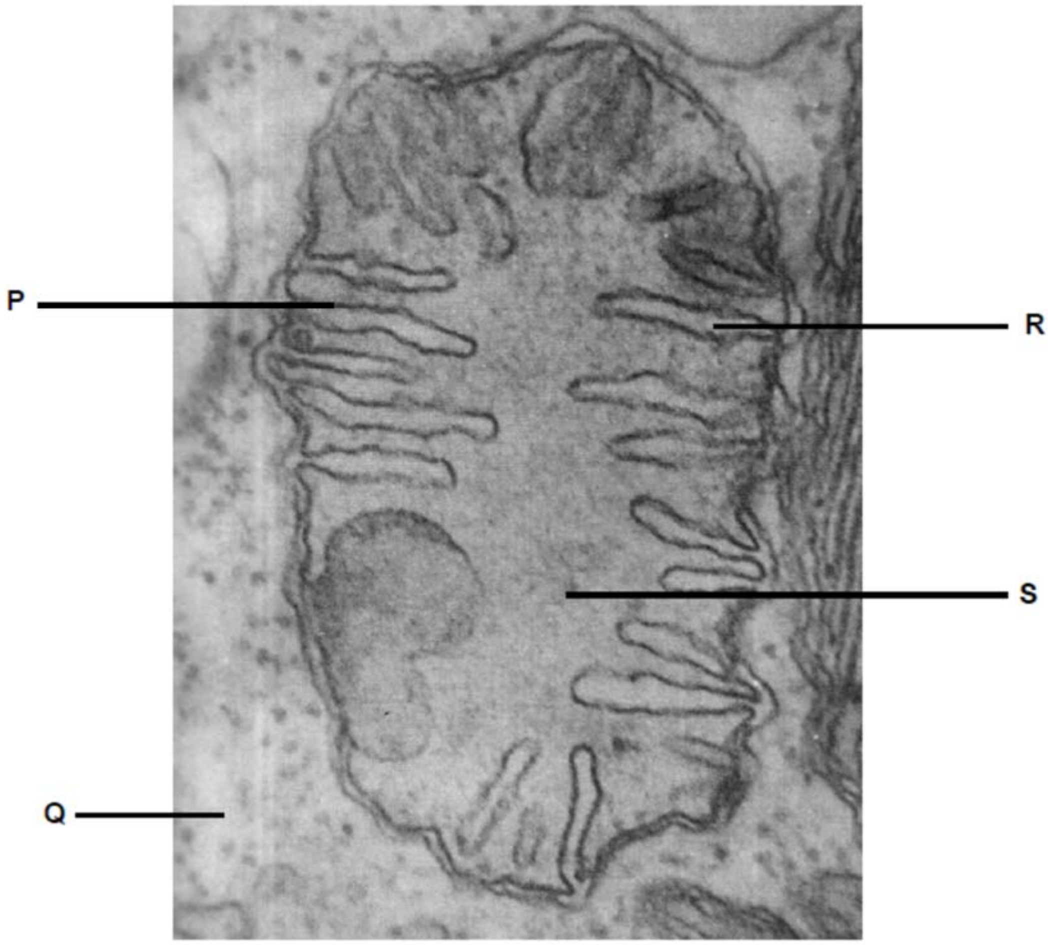
A sample of the same pond weed was exposed to four different wavelengths of light of the same intensity for the same time. The table shows the number of bubbles produced by the pond weed at each wavelength of light.

experiment	number of bubbles			mean number of bubbles
1	15	14	16	15
2	12	11	13	12
3	3	4	2	3
4	1	2	0	1

Which row shows the number of bubbles produced by the different wavelengths of light investigated?

	mean number of bubbles			
	440nm	520nm	560nm	670nm
A	1	12	15	3
B	3	1	12	15
C	12	15	3	1
D	15	3	1	12

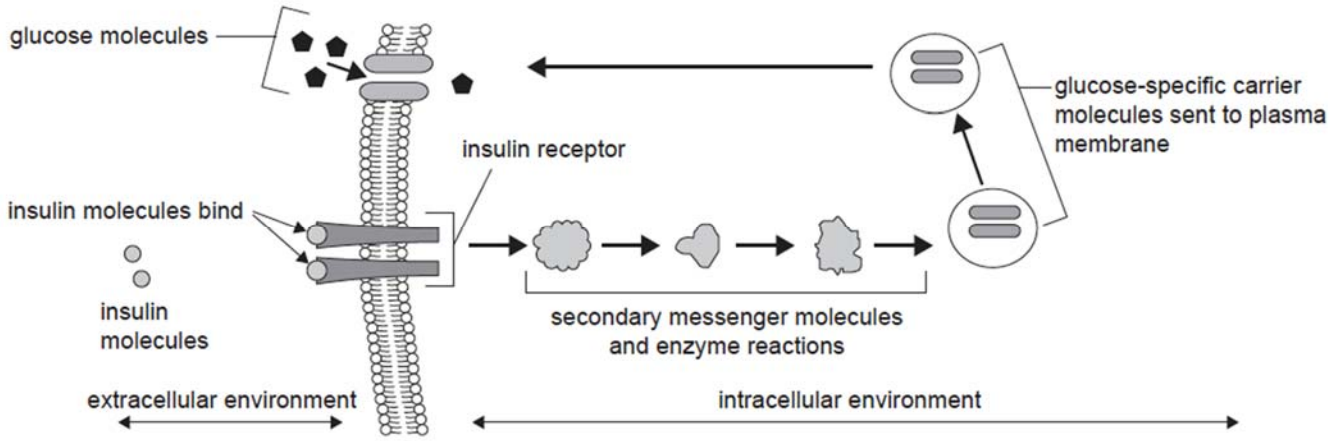
23 The figure below shows an electron micrograph of an organelle.



Match the following processes with the structures labelled P to S above.

	Breakdown of fructose-6-phosphate	Oxidative phosphorylation only	Temporary lowering of pH	Formation of water
A	Q	P	R	S
B	Q	R	S	P
C	R	S	P	Q
D	S	P	Q	R

24 The diagram below shows a summary of the steps in an insulin signalling pathway that results in increased glucose uptake.



A scientist studied the insulin signalling pathways of two female patients, Rachel and Rebecca.

Rachel's pathway is the same as that shown in the diagram above.

The scientist discovered that the gene that encodes the insulin receptor in Rebecca's has a mutation. Insulin molecules cannot bind to Rebecca's insulin receptors.

From this information, it would be correct to conclude that

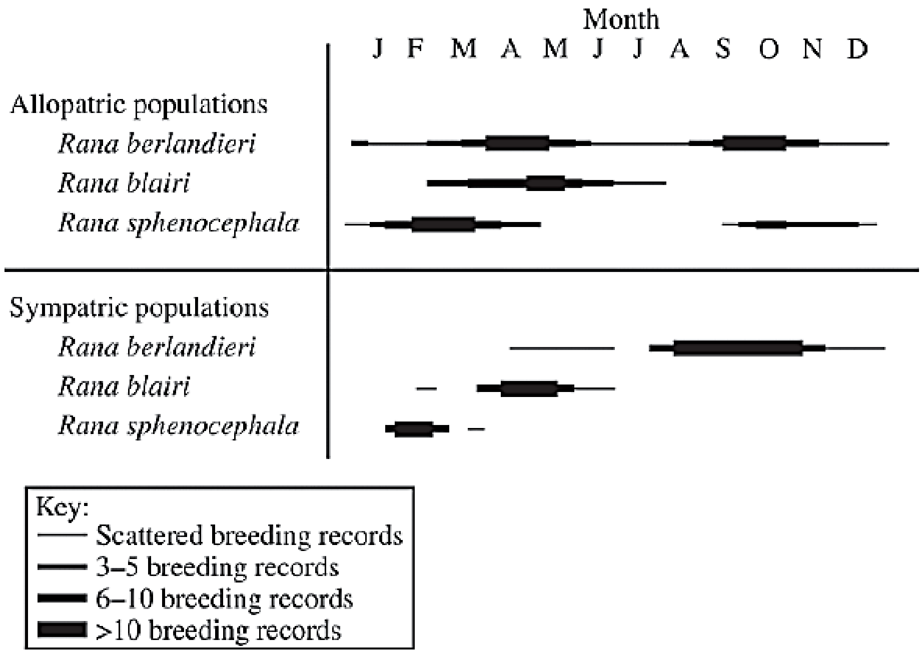
A insulin acts as a hydrophilic signalling molecule in Rachel and Rebecca.

B there would be more glucose-specific carrier molecules in Rebecca's plasma membranes than in Rachel's.

C the binding of insulin molecules to the receptor initiates transduction and the uptake of glucose into Rachel's cells.

D the presence of insulin in Rebecca would cause an increase in the concentration of the secondary messenger molecules.

25 The figure below represents a temporal plot of recorded breeding activities of both allopatric and sympatric populations of three related species of Rana (leopard frog). Rana males attract females by vocalising. The key quantifies numbers of observed breeding records for each population.



Which of the following is correct based on the data?

A Allopatric populations of all three species have largely non-overlapping breeding seasons.

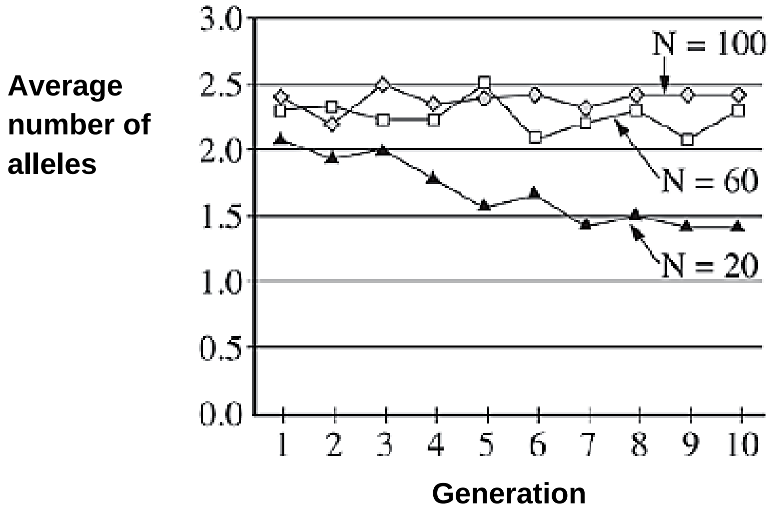
B Allopatric populations of *R. berlandieri* and sympatric populations of *R. blari* have expanded their breeding seasons compared with allopatric populations.

C Sympatric populations of *R. sphenocephala* have expanded their breeding seasons compared with allopatric populations.

D Sympatric populations of all three species have reduced the overlap of their breeding seasons compared with allopatric populations.

26 Biologists performed an experiment with flies to examine the effects of population size on the maintenance of genetic variation. From a large source population, they randomly assigned eggs to three experimental populations of size N, equal to 20, 60 and 100. For later generations they collected N eggs from each population and moved them into identical vials that contained fresh medium. They counted the number of adult flies that emerged and used tissue samples from the adults for genetic analyses.

Genetic variation was measured by scoring alleles at several polymorphic loci and expressed as the average number of alleles at those loci. The results are summarised in the figure below.



Which of the following best describes the change in genetic variation over time?

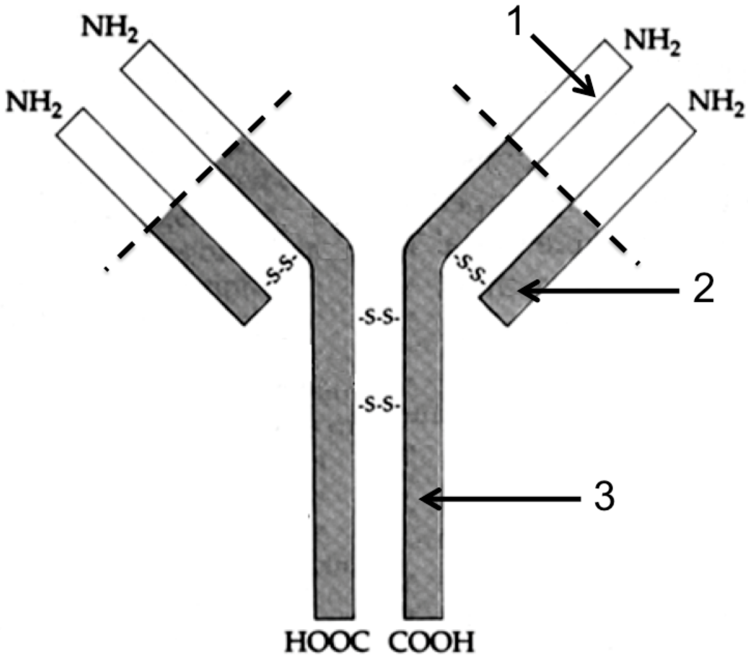
A It declined in all three populations.

B It was unchanged in the largest population and increased in the two smaller populations.

C It was unchanged in all three populations.

D It was unchanged in the two larger populations and declined in the smallest population.

27 The diagram represents the general structure of an antibody.



Which of the following numbered part(s) of the diagram represent the part of the antibody that has the same sequence of amino acids in all antibodies?

A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3 only

28 Climate change has led to increasing temperature. The environments where birds, fishes, and other marine species live have become warmer. What is not an effect of climate change on these organisms?

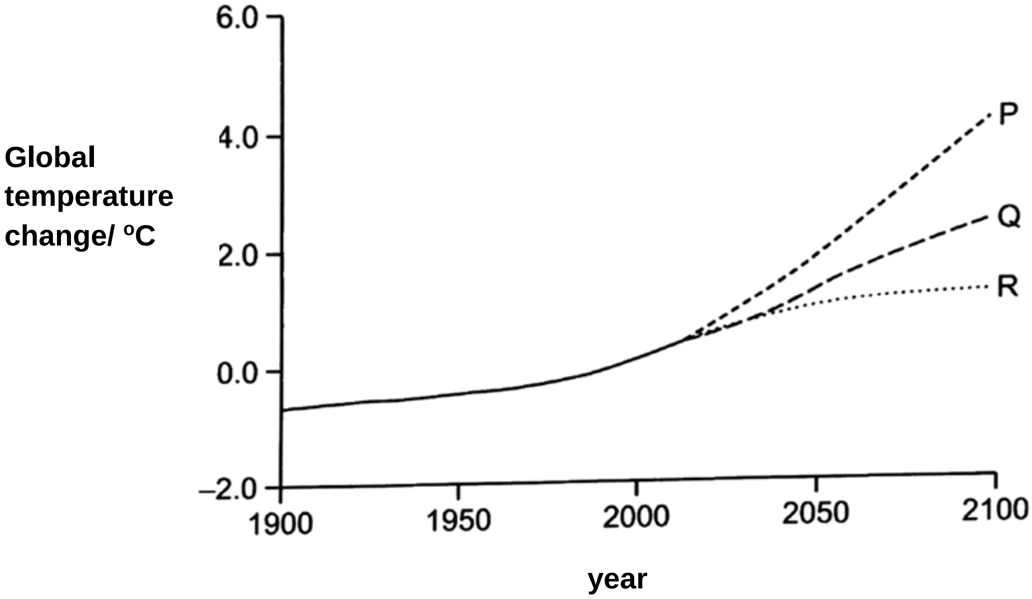
A As smaller marine prey species shift their habitats, larger predator species may follow them.

B Certain fish species migrate in response to seasonal temperature changes, moving northward or to deeper, cooler waters in the summer and migrating back during the winter.

C The birds are moving north to feed and breed in the summer, then moving further south to spend the winter in warmer areas.

D The birds follow a regular seasonal migration pattern during wintering periods.

29 The graph shows the predicted changes in global temperatures using three different models, P, Q and R. Model Q assumes that no new factors act to influence the rate of climate change.



The predictions based on models P and R can be explained using some of the following statements.

1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.

2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperature increase.

3 Rising sea temperature will cause increase growth of photosynthetic algae.

4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions P and R?

	statements that support prediction P	statements that support prediction R
A	1 and 3	2 and 4
B	1, 2 and 4	3
C	2	1, 3 and 4
D	3 and 4	1 and 2

30 Corals are among the first indicators of climate change. When ocean temperatures get too hot for too long, corals undergo a process called bleaching. Which statements are true about this process?

1 With increased levels of sediment in seawater, the zooxanthellae may lose substantial amounts of their photosynthetic pigmentation, which decreases rates of photosynthesis to result in bleaching.

2 Bleaching occurs when abnormally high sea temperatures cause corals to expel the zooxanthellae living in them.

3 Zooxanthellae are able to use the oxygen and waste materials of the host, supplying carbon dioxide and food substances in return.

4 With high sea temperatures and decreasing planktons, corals use the zooxanthellae as their food source.

A 1 and 2

B 1 and 4

C 2 and 4

D 2 and 3